

THE INTERMOUNTAIN POWER PROJECT 1600 MW HVDC TRANSMISSION SYSTEM

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Abstract

The 1600 MW HVDC transmission system of the Intermountain Power Project (IPP) is primarily designed to deliver energy from a coal-fired, two unit generating plant in Delta, Utah to load centers in Southern California. To minimize disturbance to the adjacent AC network and the relatively isolated generating plant, several special control features as well as high short time and steady-state overload capability are required in the design of the IPP HVDC converter stations.

The power system reliability requirements of the Western Systems Coordinating Council (WSCC) on the DC transmission system also resulted in the specification of guaranteed reliability and availability requirements.

The main features in the design of the main circuit equipment, HVDC controls and the reliability and availability consideration of the IPP transmission system are presented in this paper.

INTRODUCTION

The IPP, a generation and transmission project in the Western United States (Figure 1), is owned by the Intermountain Power Agency (IPA) which is a political subdivision of the state of Utah. IPA has designated the Los Angeles Department of Water and Power (LADWP) as the Project Manager and Operating Agent for IPP.

LADWP performed conceptual system studies for the Project and is responsible for the

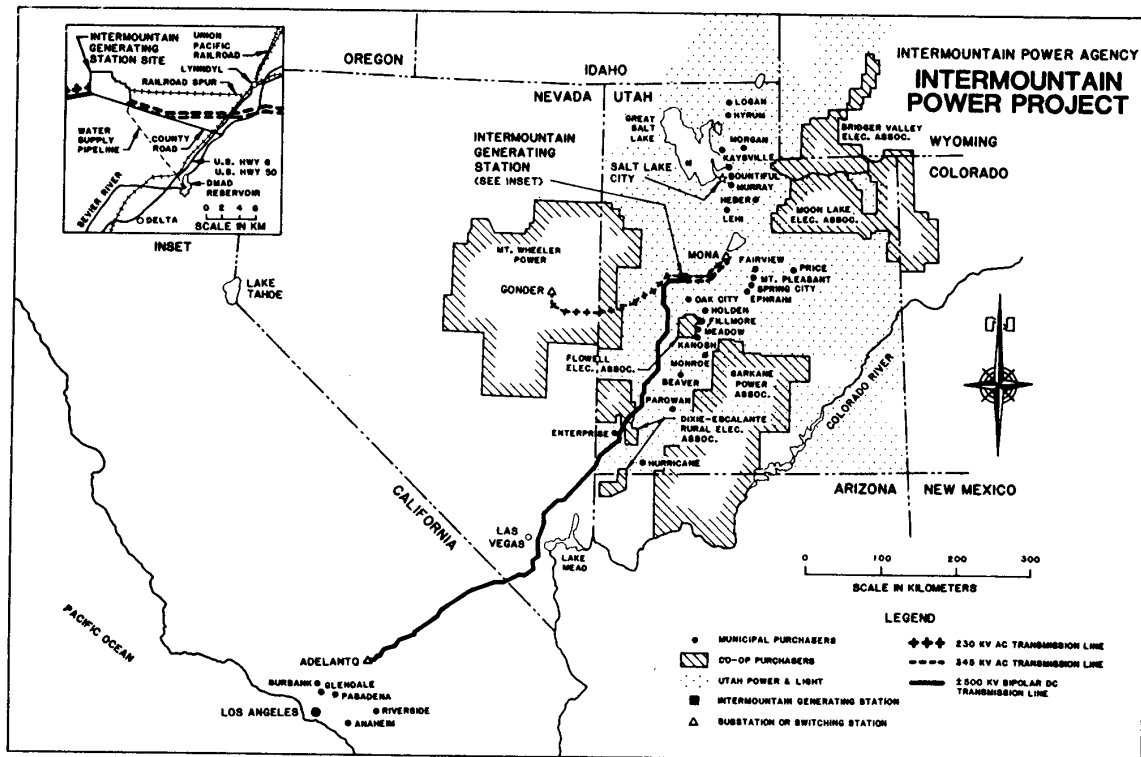


FIG. 1. INTERMOUNTAIN POWER PROJECT FACILITIES LOCATIONS

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overall project design, the managing of the construction and the integration of all the facilities for the Project.

Description of the Project has been previously presented [1]. This paper will discuss the design of the IPP HVDC Transmission System.

Since the IPP transmission system is interconnected into the LADWP, Sierra Pacific Power Company and Utah Power and Light Company ac systems (Figure 2), which are integral parts of the WSCC network in the

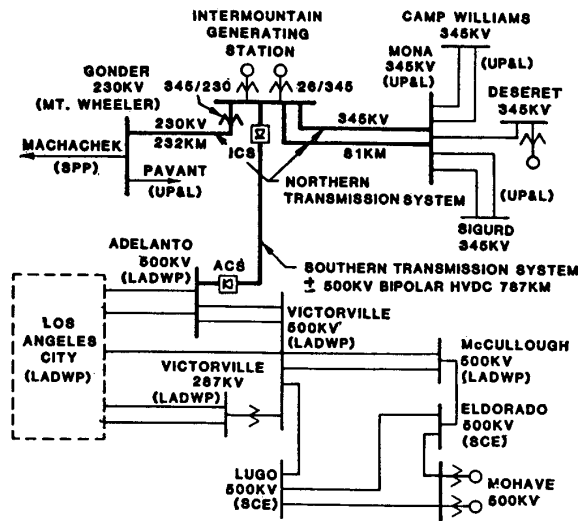


FIG. 2. IPP Transmission Systems and Interconnection Points

Western United States, transmission disturbances originating on the IPP system can impact not only its neighboring utilities through the immediate interconnections but also other utilities in the WSCC network. These interconnected operation considerations have led to the project implementation of a 2 per unit (pu) short time pole power overload for system stability enhancements and the adoption of an exceptionally low converter station bipole outage frequency of one in five years to provide the required operational reliability. These and other features have been adopted by the project with the goal of achieving a reliability level exceeding what was originally prescribed by WSCC criteria. Subsequently, the WSCC adopted a more restrictive requirement which incorporated reliability goals similar to those of IPP into its revised reliability criteria.

The project is located in a seismically active area. The Adelanto area is classified in Uniform Building Code as Zone 4 (0.5g). The area in Utah for the generator site is also seismically active, with the classification of Zone 3 (0.2g). All suppliers for the IPP are required to provide equipment designs that meet the applicable seismic design criteria.

The IPP HVDC Transmission System has both its converter stations and the entire transmission line in a desert climate. Both converter stations are at high altitude, with the Intermountain Converter Station (ICS) at 1360 m and Adelanto Converter Station (ACS) at 915 m above sea level. The weather pattern along the entire HVDC transmission system permits a steady accumulation of contaminants on insulator surfaces during months of dry period with occasional rainfall

which could create conditions susceptible to insulator flashover. This in combination with the insulator contamination problems experienced in the Sylmar Converter Station [2][3] has led to relatively long creepage distance requirement for the project.

THE HVDC TRANSMISSION LINE AND ELECTRODE SYSTEM

The +500-kV dc transmission line between the two converter stations is 785 km in length. The line extends along a route with a minimum elevation of 275 m and a maximum elevation of 1900 m above mean sea level with terrain varying from low to high desert to forest lands.

There are 1930 towers, all free standing steel lattice structures, with loading capabilities of 100 mph winds on bare conductors or one inch of radial ice on conductors with 50 mph winds. The insulation of the transmission line consists of V-string assemblies for all suspension towers and quad-bundle deadend assemblies for deadend towers. All conductor insulators have 222 kN M & E rating and are of dc design for increased performance. In general, a leakage distance of 14.7 m for suspension strings and 16.4 m for deadend strings was used; however, in designated areas of high contamination, these were increased to 19.7 m.

Air gaps to the towers are 4.0 m horizontal and 4.4 m vertical. Two shield wires at the top of the tower provide lightning protection with a shield angle of 13 degrees.

The line conductors consist of three bundled 1235 mm² Thrasher.

Both the California and the Utah overhead electrode lines consist of two 908 mm² Falcon conductors which are in the shield wire positions of the transmission line towers exiting both converter stations. The insulation of the electrode line consists of single and double string of ANSI Class 52-5 (145 x 250 mm) porcelain insulators for the suspension and deadend towers respectively.

The electrode line conductors are carried on the transmission line towers for 84.5 km from the ACS and 44 km from the ICS; from these positions, they are carried on their own poles for 6.8 km and 1.4 km respectively. The final 3.2 km of the California electrode line consists of six 507 mm² copper underground insulated cables.

Each ground electrode consists of 60 deep wells. The nominal well depth for the Utah and California electrodes are 87 meters and 70 meters respectively. Each electrode covers an area in excess of .65 km².

In monopolar ground return mode of operation, the full dc current will pass through both the ground electrodes. Therefore, the electrodes are designed for a ground current in either direction of 3700 A and then rampdown to 2600 A in 6.7 minutes, to handle the specified overload operation. The electrodes are designed to handle the 2600 A current for a continuous period of time up to 15 days. However, monopolar operation with ground current will normally last less than

one half hour since switching to the metallic return mode would be done in that time. The effects of ground current on underground metallic facilities will thereby be limited.

DESIGN OF THE CONVERTER STATIONS

Overload

With only two converters to normally carry the major part of the generator station output, special consideration had to be given to the consequences of a pole outage. The result was to design each converter with capacity to double its output if the prefault power level was below or at the rated 800 MW per pole, then ramp down at the rate of 60 MW per minute until the maximum continuous level of 1200 MW is reached. This is illustrated in Figure 3.

The said ramping speed was set so as to enable a controlled runback of the generator output to avoid unit tripping. A control signal for loss of a pole is given from the converter station to the generator station to start the generator runback.

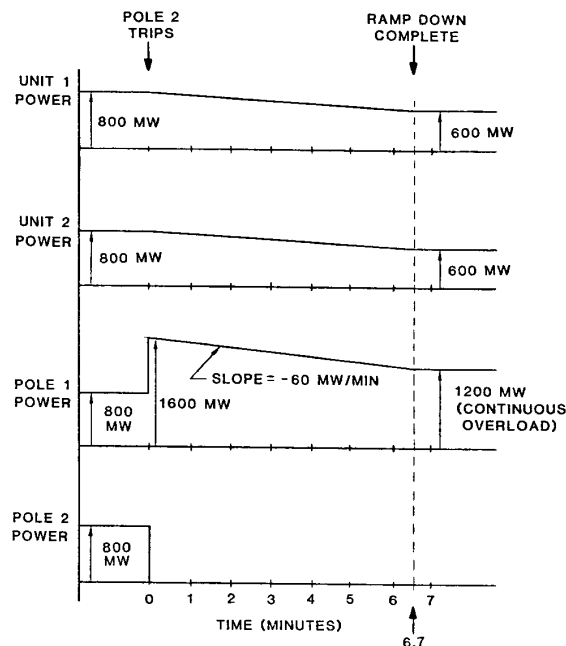


FIG. 3. IPP SYSTEM RESPONSE TO LOSS OF A POLE WHEN OPERATING IN BIPOLAR MODE

A comparison of this rather unusual power overload requirement and a 10-second 25% and 2-hour 10% current overload requirements showed the following major differences in the main circuit equipment:

- Thyristor valve redesign for 144 thyristors per single valve rather than 132.
- Increased size of the converter transformers of 7% by weight. The general design of the transformers could be retained.

- Increased size of the smoothing reactors of 11% by weight.
- Increased valve cooling capacity.

The utilization of this pole power overload capability and the resultant system stability enhancement was discussed extensively in reference 4.

Reduced DC Voltage Operation

As is well known from other HVDC projects, fog or drizzle after a long dry period during which pollution has accumulated, may reduce insulator withstand to a value below rated voltage. An operating mode with 0.7 pu line voltage (350 kV) on one or both poles is provided to make continued operation possible. The reduced voltage is accomplished by the combined use of converter transformer tap-changer operation, and increased delay and commutation angles in the respective converters. During this reduced voltage operation, the dc power of the affected pole is limited to 400 MW maximum. However, because of the continuous 1200 MW pole power overload capability, if only one pole of the bipole is affected, the system can operate in the asymmetrical bipole mode of operation and maintain 1600 MW transmission capability. The disadvantages are an increase in losses and the unbalanced ground current.

AC Filters and Reactive Power

As seen in Figure 4, there are 3 filter banks in each of the converter stations. These provide for both filtering of harmonics, and balancing of reactive power. No more than 2 filter banks are required for operation at rated bipolar load. The extra filter bank is there to increase the availability of the transmission system and to balance the reactive power at high overload. The reactive power consumption of the converters is approximately 48% of the active power at rated load.

At Adelanto, the maximum reactive power exchange with the network on the 500 kV ac bus is + or - 100 Mvar. To fulfill this criterion and at the same time meet the filtering performance requirements, specific filter sub-bank switching is controlled by a microprocessor based Reactive Power Controller (RPC) and a typical scheme is shown in Figure 5. Also, the commutation angle at Adelanto can be increased to increase the reactive power consumption by the converter which increases the hysteresis of switching and decreases the number of switching operations. This can be done in two modes. In the first mode, the dc voltage remains unaffected and the ac source voltage is increased within its limits by the tap-changer control; in the second mode, the dc voltage is lowered to provide a wider range of the tap-changer control. The design for the ICS is similar to Adelanto except the specified reactive power exchange is only + or - 50 Mvar during operation with no generators. With one or two generators in service the reactive power exchange is to be regulated by the generators.

For operation at reduced dc voltage, it

should be noted that a pole operating at 350 kV and 400 MW has approximately the same reactive power consumption as a pole at 500 kV, 800 MW. This mode could thus be met with the same size in filter MVar as for normal operation, though the filtering requirements for this mode had to be weighed into the total filter design.

telephone lines. The interference level is measured as induced voltage on a 1 km length of wire parallel to and separated by 1 km from the dc line.

The maximum allowed induced noise voltages measured on the test line are: 10 mV/km at up to 10% overload bipolar operated; 20 mV/km

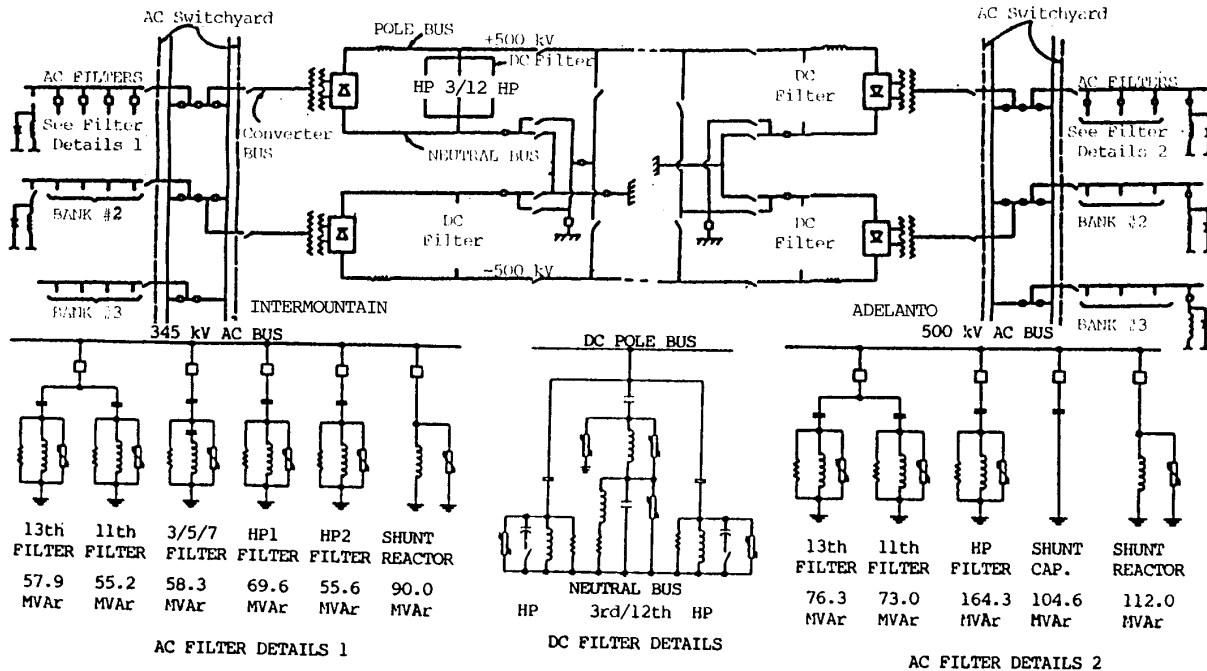


FIG. 4. SIMPLIFIED ONE LINE OF THE IPP HVDC TRANSMISSION SYSTEM

The filters have been designed for a Telephone Influence Factor (TIF) of 30, a maximum individual harmonic distortion of 1%, a total harmonic distortion of 4% and an effective harmonic distortion of 3%. The limits for harmonics fed into the generators at Intermountain are 0.7% for the arithmetic sum of 5th and 7th harmonics and 4% for the total generator harmonics.

DC Filters

Harmonic filters are used on the dc side, between the high voltage line and the neutral line of each pole, in order to reduce harmonics on the dc line to values which will not cause harmful interference in nearby

above 10% overload bipolar operation or monopolar operation up to 10% overload; and 30 mV/km with one filter branch out of service or monopolar mode of operation at above 10% overload up to 50% overload. The solution was to use 3 filter branches per pole, namely one 12th harmonic filter, and two identical filters for higher order harmonics (Figure 4). The high pass filters can be retuned by the closing of a switch to become another 12th harmonic filter; this feature is used in both filters if the 12th harmonic filter is out, and the total filtering then meets the 30 mV/km criterion. Similarly, that same criterion is also met if one of the high pass filters is out.

The 12th harmonic filter is also double tuned which provides damping for the 3rd harmonic. Sufficient damping is provided if such filter is available in one end of a pole line. The 3rd harmonic filter is provided to damp the 3rd harmonic resonance on the dc side.

Power Line Carrier (PLC) Filters

Whereas PLC is not used on ac lines connecting the ACS and the ICS, and not on the dc line, such systems are in use adjacent to the Project's lines. In particular, open wire carrier systems operating at low frequencies (10 kHz range) are in use by railroad companies. Interference limits were specified on both ac line and dc line terminals for the 30 kHz to 500 kHz range,

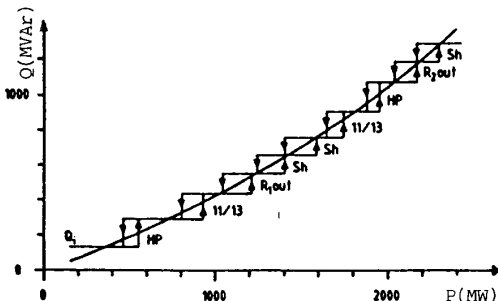


FIG. 5. TYPICAL REACTIVE POWER SWITCHING SCHEME FOR ACS IN BIPOLAR OPERATION

which led to design of filters on the feeders to converter transformers, on the high voltage dc bus, and on the neutral dc bus. In all three locations, two L-links were used; however, the dc side high pass filter provides part of one of the needed high voltage capacitors. The filters were designed to give sufficient protection also in the 10 kHz to 30 kHz range.

Metallic Return

Metallic return is an operational mode which can be used when the converters of one pole are out and the conductor of the idle pole is utilized to carry the neutral current to avoid return current through the ground.

Metallic return operation and return to normal operation are always accomplished by connecting the metallic return path and the ground path in parallel, then breaking the current in one or the other path. There are thus two breakers required for these current commutation sequences. These are called Metallic Return Transfer Breaker (MRTB) for interrupting the ground current path, and Ground Return Transfer Breaker (GRTB) for interrupting the metallic return path. The maximum currents they are designed to commute are 1.9 kA and 0.9 kA, respectively, when the converters are operating at 2560 A.

The principle design of these breakers was earlier described [5]. The interrupting units used for IPP are of oil minimum type.

Reliability and Availability

The specification for the IPP HVDC converter stations requires the following reliability and availability guarantees:

FOPCA (Forced Outage Pole Capacity Availability)	99.99 %
SOPCA (Scheduled Outage Pole Capacity Availability)	100 %
FOBCA (Forced Outage Bipole Capacity Availability)	99.0 %
SORCA (Scheduled Outage Bipole Capacity Availability)	97.0 %
PFOR (Pole Forced Outage Rate) per year	10
BFOR (Bipole Forced Outage Rate) per year	0.2

To be able to meet these design guarantees, the following measures were taken:

In order to minimize the impact on reliability and availability from the control and protection system, all subsystems that are essential for the transmission of power have been duplicated and connected in a redundant scheme. Each of the two redundant control systems has its own set of measuring transducers. The redundancy of the control system should eliminate component failures within the control and protection system as a cause of outage. Further, maintenance of the control and protection system can be done on the standby system with the HVDC system in operation.

The demand that SOPCA (Scheduled Outage Pole Capacity Availability) shall be 100% was met by a separate disconnector from each pole neutral bus to the metallic return bus and by the introduction of a breaker, called NBGS (Neutral Bus Grounding Switch), that can connect and disconnect the metallic return bus to and from the station groundmat in the ACS. No dc current will flow into the station groundmat because the NBGS will be closed only when the HVDC system is operating in the metallic return mode of operation.

Of the three ac filter banks in each station, one complete filter bank or the equivalent set of subbanks from more than one filter bank will be sufficient from a thermal rating point of view for the operation of the complete bipole up to rated power. This redundancy of the ac filters should minimize the ac filters as a cause of forced outages, and the scheduled maintenance of the ac filter banks may be performed on one bank at a time without loss of power transfer capability.

Redundancy also is provided in the dc filters. This makes it possible to maintain the dc filters with minimum interruption on HVDC system operation by taking one filter branch at a time in a station out of operation for preventive or corrective maintenance.

The auxiliary power system and the cooling systems have been provided with redundancy for all vital functions, and the major parts of these systems can be maintained during normal HVDC system operation.

Special Control Features

The control system is a fully electronic system using microprocessors. In each station approximately 200 microprocessors are used. The vital parts of the control equipment are duplicated with an automatic switchover scheme between the redundant systems in case of a failure. Among the control functions the following features are of special interest:

- Reactive Power Control
- DC Power Schedule Calculator
- Generator Runback Control
- DC Power Modulation
- Overload Recovery Control
- Electrode Current Balancing
- Isolated Startup

Reactive Power Control (RPC)

The purposes of the RPC are:

- To determine number and type of filter sub-banks needed to fulfill the ac filtering performance
- To determine the needed reactive power generation in order either to balance the reactive power, or to control the ac bus voltage
- To prevent dynamic overvoltages when the converters are blocked

- To prevent self-excitation of the IPP generators at ICS
- To execute switching action or - depending on the selected mode - annunciate to the operator that switching is due

The RPC at Adelanto also contains the program for increasing the hysteresis in switching by increasing the commutation margin.

At Intermountain, the RPC can perform the above functions if there are no generators connected. If generators are connected, the generating station controls the ac voltage and the RPC supervises the reactive power flow in order to prevent self-excitation of the generators.

DC Power Schedule Calculator (DCPSC)

The DCPSC receives information as to the portion of the net generation at Intermountain which shall be transmitted on the HVDC line, and also the scheduled power exchange with the Utah network. From these data a power order is calculated which can be selected to automatically control power on the HVDC line.

Generator Runback Control (GRC)

In case of a pole outage the GRC will automatically order the generators at Intermountain to reduce their power with a predetermined ramping function. The short time pole power overload capability is fully utilized during such an operation.

DC Power Modulation

A set of control functions can modulate the dc power by means of additional additive inputs to the operator-set power order.

- The Frequency Control consists of a proportional controller on pole level and an integrating control (Constant Frequency Control) on bipole level. This function is to stabilize the frequency of the generators at Intermountain during islanded operation. It should also reduce the transmitted dc power if one IPP generator is lost, or increase the power on one pole if the other pole is lost.
- The subsynchronous damping controller counteracts oscillations in the Intermountain generator turbine shafts that may develop during operation with the converter station by modulating the dc power [6].
- Six additional power modulating inputs are available for future use, three at ICS and three at ACS.

Overload Recovery Control

The Overload Recovery Control reacts on faults in the connected ac systems by ordering an additional current pulse to enable recovery to the ordered overload.

Electrode Current Balancing

A small electrode line current may flow in balanced bipolar operation due to differences in the control equipment for the poles. The electrode current balancing provides a means to counteract those differences by giving corrective current orders to both poles.

Isolated Startup

At isolated startup, the generators in Intermountain can be started with the dc transmission as the only load with minimum bump to the connected ac system. This is accomplished by operating the converter station at below the normal 10% minimum current.

Creepage Distance Selection

Environmental reasons and operating experiences from the Sylmar Converter Station led to the following minimum creepage distances on the dc side for both sites:

- Wall bushings 48 mm/kV
- Transformer bushings 48 mm/kV
- Other equipment bushings 40 mm/kV
- Suspension and post insulators 32 mm/kV

For the ac side, 28 mm/kV was used for ICS and 29 mm/kV for ACS.

Seismic Design

The seismic design accelerations specified were for horizontal movements 0.5 g for ACS, and 0.2 g for ICS. For vertical acceleration, 60% thereof applies.

These seismic criteria have led to the following special designs:

- Suspended thyristor valves
- Flexible joints (gimbals) for wall bushings and converter transformer bushings
- Use of friction spring elements between some equipment and their supporting steel rack or pedestal
- Flexible joints between equipment
- Structurally reinforced control cubicles

MAJOR COMPONENTS

Thyristor Valves

The converters are of 12-pulse type. The 12 electrical valves are configured in three quadruple valve assemblies. Each single valve consists of 12 valve modules each of which contains 12 thyristor levels in series. The thyristors have a wafer area of 45 cm² and are water cooled. The built-in redundancy for each valve is five thyristor levels which fulfills the specified three percent redundancy requirement. The guaranteed thyristor failure rate including thyristor electronics is 0.55 percent per year.

Each thyristor level contains one thyristor, a saturable reactor, voltage grading circuit and a thyristor control unit.

The 3 quadruple valves, with a weight of 51 tons each including the arresters, are suspended from the valve hall ceiling because of the seismic design requirements. The suspension structure provides both mechanical support and electrical isolation of the high voltage (500 kV) terminal of the quadruple valve from the ceiling which is at ground potential. The maximum deflection at the lower end of the quadruple valve for the specified seismic condition at ACS is ± 750 mm. Umbilicals - water cooling circuits and light guides - and electrical connections have been specially designed to allow the needed movement at the different connections. Telescopic tubes with flexible braids inside are used for the electrical connections.

Details of the design requirements and the testing program for the thyristor valves were described in Reference 7.

Converter Transformers

The converter transformers are single phase units with two valve windings for the wye connected and the delta connected bridges respectively. Main data are:

		ICS	ACS
Rated Voltage Line side	kV	$345/\sqrt{3}$	$500/\sqrt{3}$
Rated Voltage Valve side	kV	$207/\sqrt{3}$	$202/\sqrt{3}$
Rated power	MVA	312	305
Tap Changer range	%	+12 -6 X 1.25	+16 -6 X 1.25
Impedance	%	13.0	12.6
Continuous overload power p.u.	p.u.	1.5	1.5
Standard impulse voltage test-line side	kV	900	1300
-valve side	kV	1550/1050	1550/1050
Switching Impulse voltage test -line winding	KV	845	1080
-valve winding	kV	1290/870	1290/870
Losses, 800 MW base %		0.32	0.32

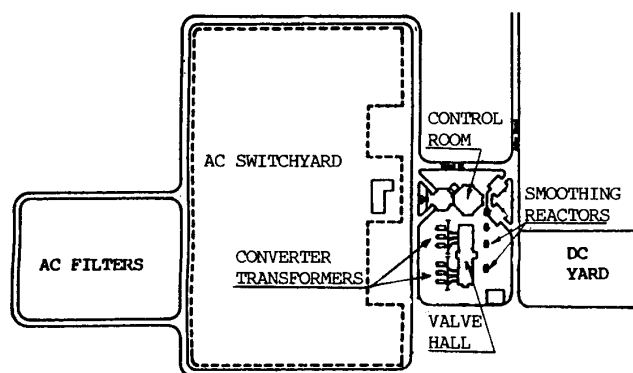


FIG. 6. ADELANTO STATION LAYOUT

The tank with its core and coil assembly had a shipping weight of 270 tons, and the fully assembled transformer weighs 430 tons.

Filter Subbank Switchers

In investigation of load rejection cases it was determined that under worst case scenarios, there is a need to disconnect filters at high overvoltages. The switchers thus had to be qualified for disconnecting capacitive load with transient recovery voltage of 1072 kV for ICS, and 1300 kV for ACS. Conventional SF6 breakers with four interrupting elements were modified for this purpose.

Layout

The general layout for Adelanto is as shown in Figure 6. Intermountain is similar in arrangement but a mirror image. The main areas have the following approximate sizes:

Area	ACS, m ²	ICS, m ²
AC filters	62,000	66,000
AC switchyard	130,000	60,000
Converter yard	14,500	14,500
DC yard	27,000	27,000

The valve hall dimensions are: 30 m long 20 m wide and 23 m high inside.

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Discussion

S. C. Kapoor (General Electric Co., Malvern, PA): The authors of this paper have provided an interesting arrangement of the dc filters where the high-pass filter can be converted into a 12th harmonic tuned filter in the event of loss of the main 12th harmonic filter. It is stated that a 3rd harmonic filter is required to damp the 3rd harmonic resonance, and that sufficient damping is provided if such a filter is available in one end of a pole line. One would expect that the resonance would be seen from either end, and that the filters would be required at both ends if the resonance was a concern. Would the authors clarify how only one filter provides sufficient damping under normal bipolar and monopolar conditions, or is the problem related to some abnormal condition such as bypass, etc.? Would the authors also state the problems experienced with the 3rd harmonic resonance that led to the need of such a filter? Are any comparative results available with and without the 3rd harmonic filter?

The authors are to be congratulated for providing useful, technical information on such a large, complex HVDC project.

Manuscript received February 26, 1987.

J. S. McConnach (Ontario Hydro, Toronto, ON, Canada): The authors have given some general indications of the special measures adopted to achieve the desired reliability and availability. Can they please be more specific in identifying the equipments duplicated, with particular reference to the control and protection systems and subsystems and their auxiliaries and also the auxiliary power system and cooling systems? Also are any statistics available on the operating experience to date relating to pole and bipolar forced outages?

Presumably a reliability design study was carried out as part of the design stage of the project and I would encourage the parties involved to consider a paper on this subject.

Finally I have two further questions for the authors.

1) What is the estimated incremental cost for the pole overload feature expressed as a percentage of the total converter terminal costs?

2) What are the converter terminal losses (per terminal) for the project under normal bipolar operation at rated power expressed as a percentage of rated power of 1600 MW?

Manuscript received March 2, 1987.

Carson W. Taylor (Bonneville Power Administration, Portland, OR): The paper discusses dc power modulation. Recently, however, there has also been interest in extinction angle (gamma) modulation. Gamma modulation was selected for the Itaipu project [A]. Studies show that gamma modulation at Adelanto should improve system dynamic performance [B], [C].

Could the authors discuss the feasibility of gamma modulation on the IPP dc link? In particular, what steady-state extinction angle would be recommended, and what would be the modulation limits?

For inverter end ac system stability (including voltage stability), either power/current modulation or gamma modulation could be used. In comparing these alternatives, I request the authors' viewpoints with regard to telemetry requirements, steady-state losses, and steady-state reactive requirements. Would dc modulation be cost-effective compared to static var compensators proposed for the Pacific Southwest?

The authors provide valuable information on the equipment changes to provide additional overload capability. For new system preliminary planning purposes, could this be roughly translated into a percent increase in converter dollars/kW cost for a 1600-MW 500-kV project?

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[C] C. W. Taylor, discussion of reference [B] (to be published in *IEEE Transactions on Power Delivery*).

Manuscript received March 2, 1987.

C. T. Wu and L. Engstrom: The authors thank the discussers for their interest in the paper.

Mr. S. C. Kapoor's questions are related to the third harmonic filter on the dc side. During the design study of the dc filter, it was established that during certain operating conditions, the combination of the dc transmission line, smoothing reactor, dc filter capacitance, and the ac impedance reflected to the dc side has resonant points near the third harmonic. Since the contract required that the dc system be designed to avoid resonance near first, second, and third harmonics, the third harmonic dc filter was provided to meet the contract requirement. The calculation also showed that with the third harmonic filter at one end only, there is sufficient change in the impedance as seen from either end of the line to avoid resonance.

The effect of the third harmonic dc filter was discussed in detail in a paper titled "DC System Resonance Analysis," authored by Mr. M. P. Bahrman, et al. and published in the *IEEE TRANS. POWER DEL.*, Vol. PWRD-2, No. 1, January 1987 and will not be further elaborated here.

Mr. J. S. McConnach requested more specific information on the equipment/system redundancy that was provided for the IPP HVDC system. The basic redundancy requirement can best be summarized as:

"... under normal operating conditions, no single failure of equipment (within the converter station) shall cause a forced outage resulting in a DC power transfer capacity reduction greater than 800 MW and no combination of failures of equipment ... within a pole shall cause a forced outage resulting in a DC power capacity reduction greater than 800 MW."

To meet this requirement, all equipment within the converter stations is segregated to a pole basis. This applies to the control and protection systems, subsystems and their auxiliaries, auxiliary power systems, cooling systems, and main circuit equipment. For the control system, redundancy is accomplished by following three principle schemes: duplication of functions, back-up system, and distribution of functions into three control hierarchies: bipole, pole, and converter levels. The converter control represents the lowest level of control.

The only area that cannot be segregated into a pole basis is the bipole control and protection. However, the dc system can continue to operate, at a degraded level of performance, without most of the bipole functions via manual operation.

The equipment on the converter control and bipole control levels is duplicated with an active and a standby system. The standby system is capable of immediate takeover should the active system stall. The pole control functions are not duplicated because the pole is redundant for a bipole system and the converter control can serve as a low-level back up.

Statistics on pole and bipole outages are being kept. A paper on the first year operating experience that will provide outage statistics is under preparation.

The incremental cost for the overload features provided for the IPP system is in the same order as the incremental changes that are required for the main circuit equipment that were detailed in the paper. The converter terminal losses under normal bipolar operation at rated power should be approximately 0.61 percent and 0.62 percent of the rated power for the Adelanto and Intermountain terminals, respectively.

Mr. C. W. Taylor's questions are concerned with dc power modulation. In particular, Mr. Taylor is interested in the gamma modulation for ac system stability enhancement. To respond fully to the questions raised required a paper in its own and is beyond the scope of this paper. The study of gamma modulation at Adelanto referenced by Mr. Taylor concerns a proposed multiterminal system (Adelanto–Mead–Phoenix) which is not part of the IPP system. We are not in a position to provide better information than what is available in the reference [B] cited by Mr. Taylor.

The IPP HVDC system does have local voltage control capability at the inverter terminal when it is operating in the increased reactive power consumption mode. This is accomplished by increasing the extinction angle to increase the reactive power consumption of the converter. This function is for steady-state local voltage control but not the gamma modulation for stability enhancement.

The incremental cost to provide the overload capability is addressed in our response to Mr. McConnach.

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